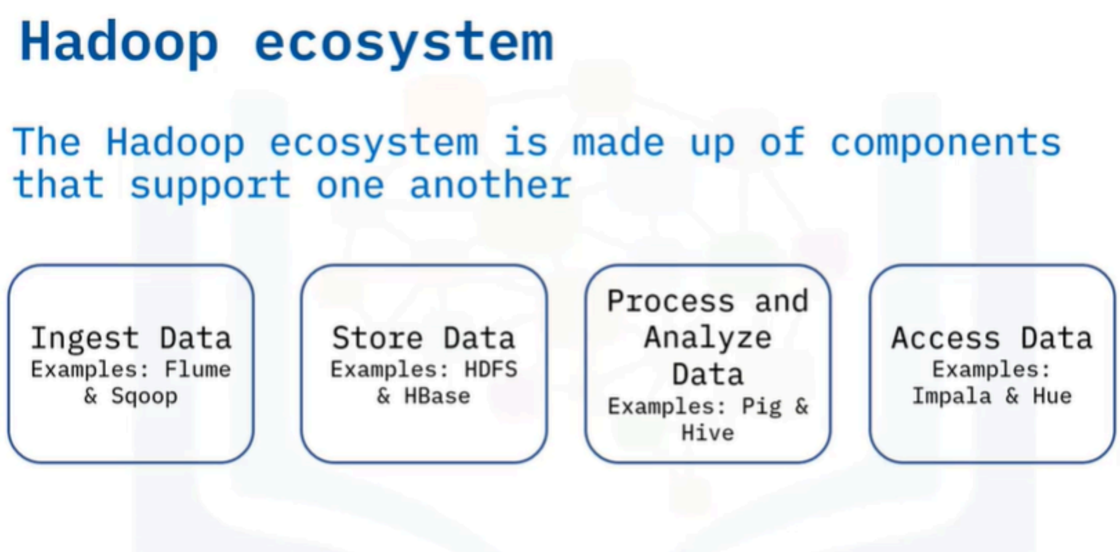
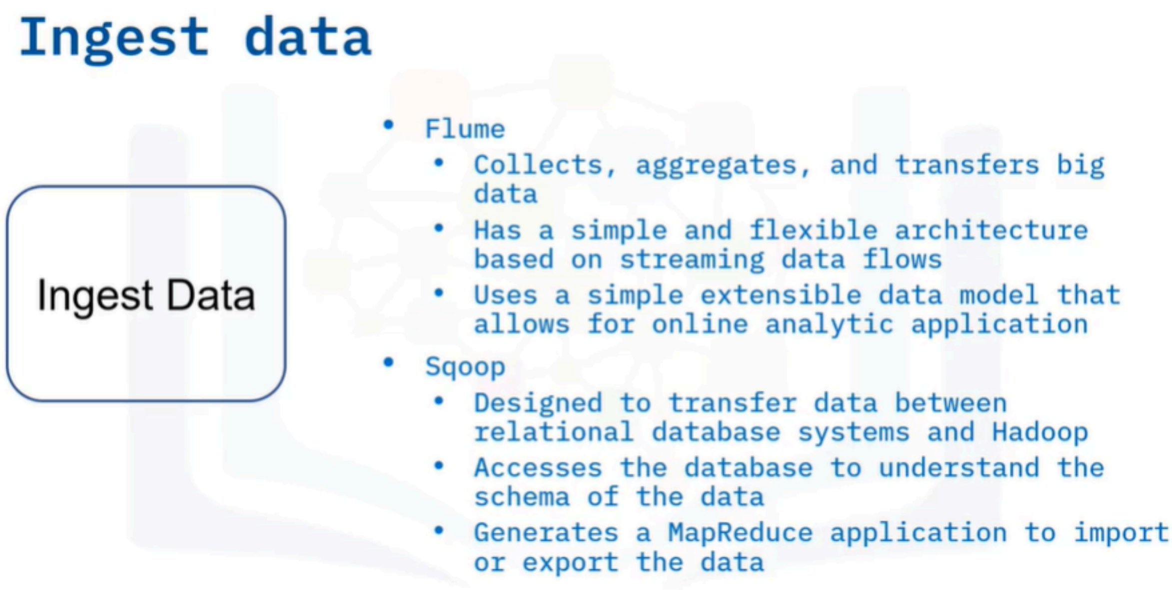
Hadoop Ecosystem

Sure! Let's dive into the Hadoop Ecosystem, which is a powerful framework for processing large amounts of data, often referred to as Big Data.

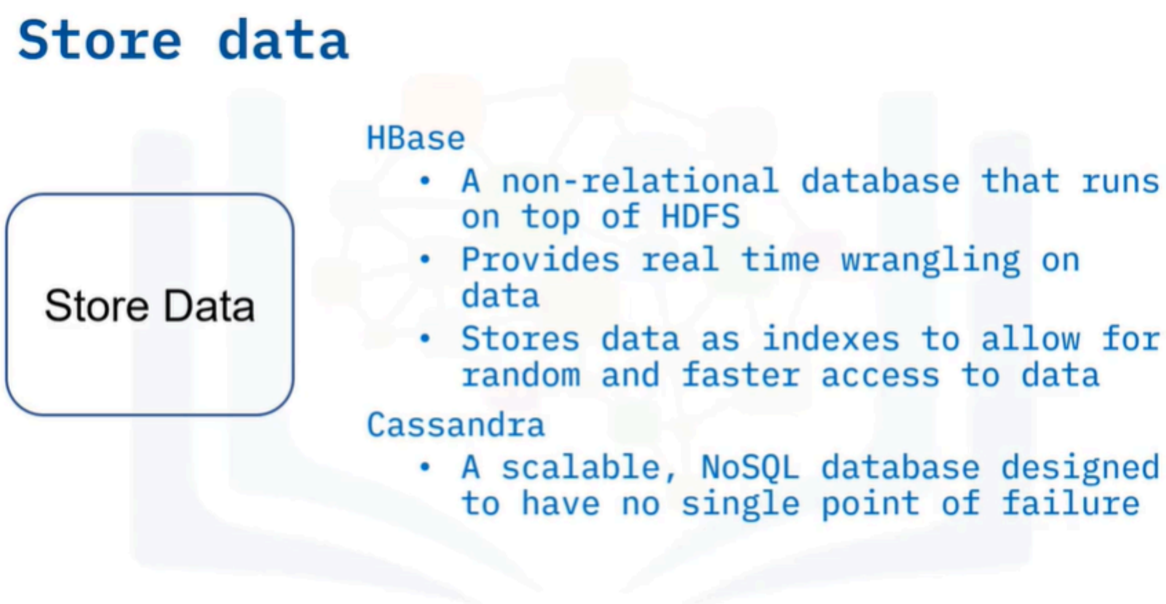
At its core, the Hadoop Ecosystem consists of several stages that work together to handle data efficiently. Think of it like a factory assembly line. First, data is collected from various sources, which is called "Ingest." Tools like Flume and Sqoop help gather this data and send it to storage systems like HDFS (Hadoop Distributed File System). Next, the data is stored and organized, making it ready for processing. After that, the "Process and Analyze" stage kicks in, where tools like Pig and Hive help analyze the data, similar to how a chef prepares ingredients for a delicious meal. Finally, in the "Access" stage, users can retrieve and interact with the refined data using tools like Impala and Hue, much like serving the finished dish to guests.



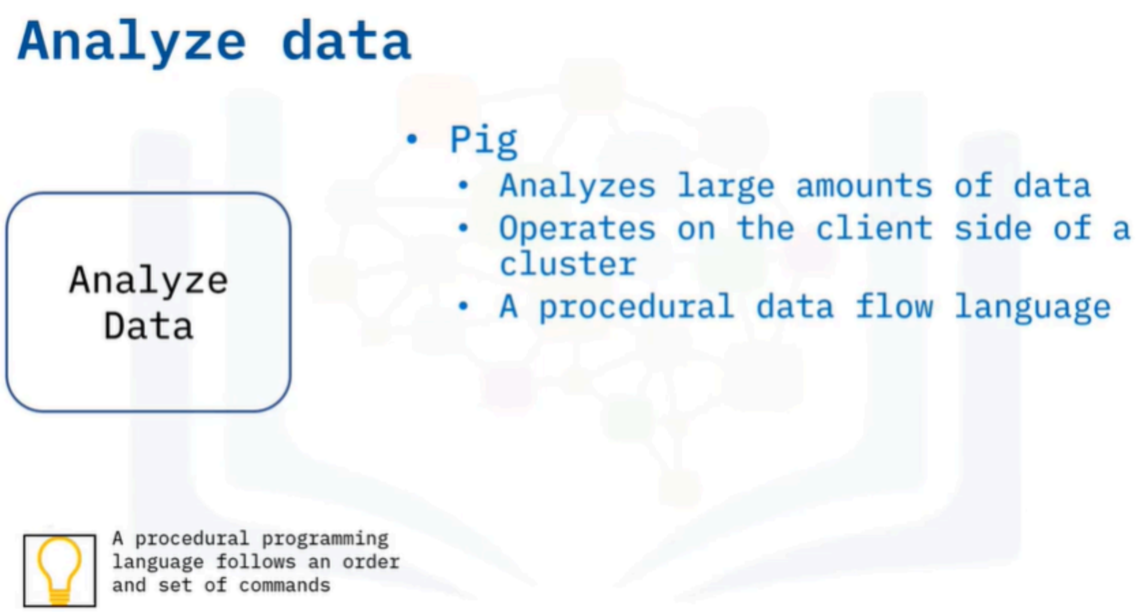
- Apart from the core components of Hadoop, Hadoop Common refers to the common utilities and libraries that support the other Hadoop modules.
- Hadoop Distributed File System (HDFS) stores the data collected from the ingestion and distributes the data across multiple nodes.
- MapReduce is used for making Big Data manageable by processing them in clusters.
- Yet Another Resource Negotiator (YARN) is the resource manager across clusters.
- The extended Hadoop Ecosystem consists of libraries or software packages that are commonly used with or installed on top of the Hadoop core.
- The Hadoop ecosystem is made up of components that support one another for Big Data processing. We can examine the Hadoop ecosystem based on the various stages. When data is received from multiple sources, Flume and Sqoop are responsible for ingesting the data and transferring them to the Storage component, HDFS and HBase.
- Then, the data is distributed to a MapReduce framework like Pig and Hive to process and analyze the data, and the processing is done by parallel computing. After all that is done, tools like Hue are used to access the refined data.

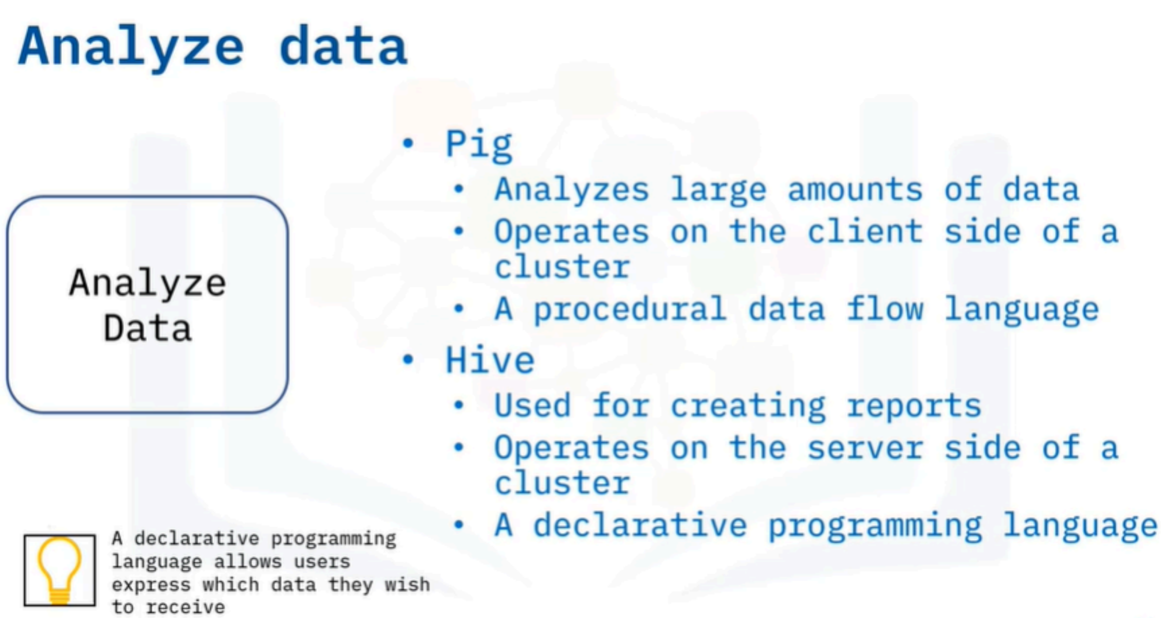


- Ingesting is the first stage of Big Data processing. Whenever you deal with big data, you get data from different sources. You then use tools like Flume and Sqoop. Flume is a distributed service that collects, aggregates, and transfers Big Data to the storage system.
- Flume has a simple and flexible architecture based on streaming data flows and uses a simple extensible data model that allows for online analytic application.
- Sqoop is an open-source product designed to transfer bulk data between relational database systems and Hadoop. Sqoop looks in the relational database and summarizes the schema. It then generates MapReduce code to import and export data as needed. Sqoop allows you to quickly develop any other MapReduce applications that use the records that Sqoop stored into HDFS.



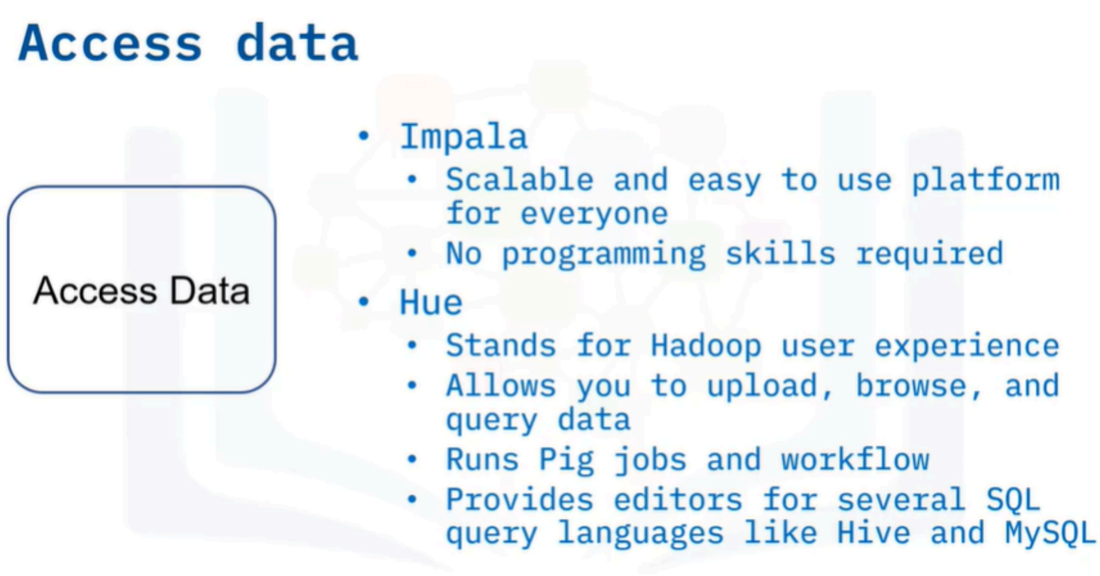
- HBase is a column-oriented non-relational database system that runs on top of HDFS.
- It provides real time wrangling access to the Hadoop file system. HBase uses hash tables to store data in indexes and allow for random access of data, which makes lookups faster.
- Cassandra is a scalable, NoSQL database designed to have no single point of failure.





- In the Analyze Data stage, Pig is used for analyzing large amounts of data. Pig is a procedural data flow language and a procedural programming language that follows an order and set of commands.
- Hive is used mainly for creating reports and operates on the server side of a cluster. Hive is a declarative programming language, which means it allows

users to express which data they wish to receive.



- The final stage is “Access data,” where users have access to the analyzed and refined data. At this stage tools like Impala are often used. Impala is a scalable system that allows non-technical users to search and access the data in Hadoop. You do not need to be skilled in programming to use Impala. And Hue is another tool of choice at this stage. Hue is an acronym for Hadoop user experience. Hue allows you to upload, browse, and query data. You can run Pig jobs and workflow in Hue. Hue also provides a SQL editor for several query languages like Hive, and MySQL.

Summary

In this video, you learned:

- The four main stages of the Hadoop Ecosystem are ingest, store, process and analyze, and access
- Some examples of the tools associated with each stage
- How each stage interacts with the others